



COURSE ORIENTATION

Big Data Analytics in Finance

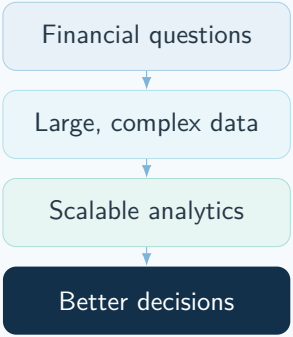
Course structure, learning journey, assessment, and expectations

Dr. Nguyen Tuan Long National Economics University | 2026

Welcome to the Course

Finance is becoming a data-intensive, real-time decision discipline.

- Financial institutions process transactions, markets, customers, documents, and alternative data.
- Modern analysis must combine financial reasoning with scalable computing.
- The course connects Python analytics, machine learning, and big-data platforms.



Course promise

Move from small, local notebooks to reproducible analytics pipelines that can process financial data at scale.

Course at a Glance

7
Core Lecture Modules

150
Minutes per session

1
Capstone Final Project

Learning format

The theory–practice balance changes by topic. Architecture-heavy sessions emphasize concepts; application sessions maximize coding time.

Course destination

A team-based final project integrating data preparation, scalable processing, machine learning, and financial interpretation.

What You Will Learn

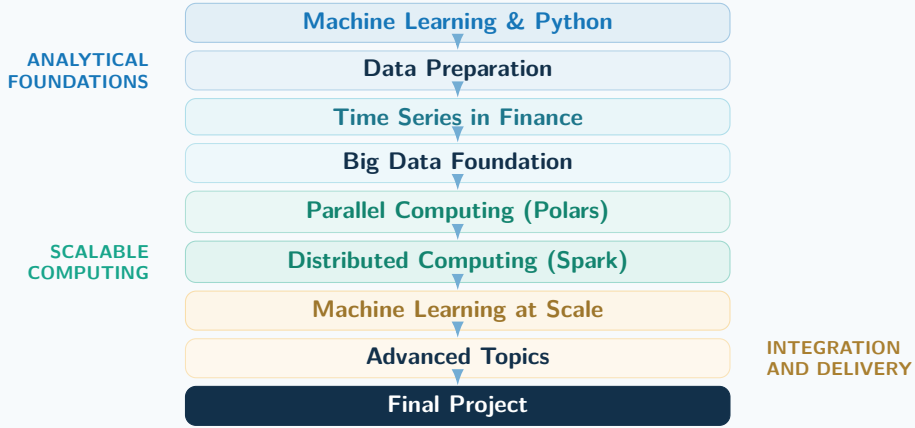
Analyze	Engineer	Evaluate
▶ Financial returns and volatility ▶ Credit and fraud data ▶ Customer and alternative data	▶ Clean and transform data ▶ Build reproducible pipelines ▶ Work with Parquet and Spark	▶ Compare models responsibly ▶ Recognize leakage and bias ▶ Communicate business value

Technical scale matters only when it improves a financial decision.

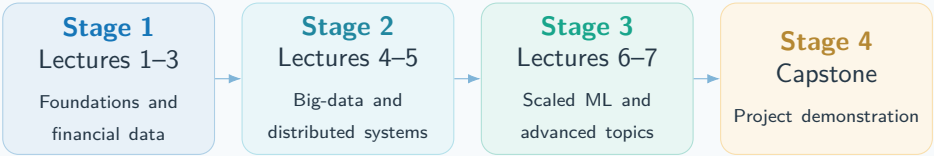
Seven-Lecture Course Architecture

Lecture	Theme	Core capability
L1	ML and Python Review	Core models, evaluation, clustering, and Scikit-learn workflow
L2	Data Preparation	Cleaning, transformation, imbalance, and feature selection
L3	Financial Time Series	Returns, volatility, windows, and point-in-time data quality
L4	Big Data Foundations	5V, cloud infrastructure, parallelism, Polars, and lazy evaluation
L5	Distributed Spark	Architecture, DataFrames, SQL, joins, shuffle, and windows
L6	Machine Learning at Scale	PySpark preparation, MLlib pipelines, and distributed evaluation
L7	Advanced Topics	End-to-end systems, alternative data, ethics, and emerging AI

Course Flow: From Foundations to Delivery



Course Journey: Four Progressive Stages



Learning logic

Each stage reuses earlier concepts at a larger data scale and with stronger requirements for reproducibility, governance, and interpretation.

Lecture 1: Machine Learning and Python Review

Part 1: Core concepts and models

- ML workflow and train–test separation
- Overfitting and underfitting
- Linear/Logistic Regression, KNN, Decision Tree, Random Forest

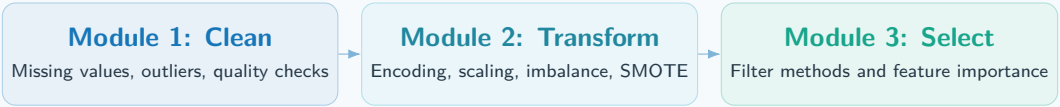
Part 2: Unsupervised learning and labs

- K-Means and elbow method
- California housing
- Tumor diagnosis
- Customer RFM segmentation

Purpose

Establish a common Python and machine-learning foundation before moving to data engineering and scale.

Lecture 2: Data Preparation & Feature Engineering



Applied lab

Clean and prepare a credit-risk dataset for modeling.

Performance insight

Demonstrate where Pandas becomes a processing bottleneck and why scale changes tool choice.

Lecture 3: Financial Time Series

- Compute returns and analyze volatility.
- Create lag, rolling, and moving-average features.
- Understand why time order changes validation design.
- Address corporate actions and point-in-time data.



Financial data principle

A technically correct calculation may still be invalid if it uses information that was unavailable at the decision time.

Midterm 1 and the Big-Data Transition

First 75 minutes

MIDTERM 1

Written examination covering Lectures 1–3.

- ML and Python review
- Data preparation
- Financial time series

Second half: Lecture 4 begins

- The 5V framework
- Cloud infrastructure
- Databricks overview
- Why single-node workflows stop scaling

Turning point

The course moves from analytical models to systems that can execute those models on large datasets.

Lectures 4–5: From One Machine to a Cluster

Module	Focus	Topics and practice
L04	Efficient single-node computing	Parallelism, vectorization, Polars, lazy evaluation, and a GB-scale processing lab
L05a	Spark foundations	Master–worker architecture, distributed computing, SparkSession, and DataFrames
L05b	Spark SQL and advanced operations	SQL, joins, shuffle, window functions, and large-scale querying practice

Core systems question

What work can be parallelized, where does data move, and which operations force expensive communication?

Lecture 6: Machine Learning at Scale

Part 1: Build the pipeline

- Clean and transform data with PySpark
- Assemble features with `VectorAssembler`
- Train models using Spark MLlib
- Connect preprocessing and modeling stages

Part 2: Evaluate at scale

- Downsampling strategies
- Distributed evaluation
- Fraud-detection baseline
- Performance and accuracy trade-offs



Lecture 7: Advanced Topics

Part 1

End-to-end pipeline

- Parquet → Spark ETL
- Scikit-learn / MLlib
- Full fraud lab
- News sentiment

Part 2

Midterm 2 + AI risk

- Written exam: 75 minutes
- Fair lending
- Data leakage
- Model risk

Part 3

Emerging directions

- LLMs in finance
- FinBERT and TimeGPT
- Synthetic data
- Review and Q&A

Final Project Presentations & Defense

Team presentation

Explain the financial problem, data, architecture, modeling decisions, results, limitations, and recommended action.

Required submission

Source code and sufficient documentation to reproduce the main analytical workflow.

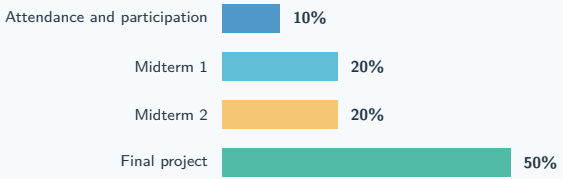
Q&A and defense

Teams should be prepared to justify:

- Data quality decisions
- Tool and architecture choices
- Evaluation metrics
- Ethical and model risks

The final product is an evidence-based financial recommendation, not only a model.

Assessment Structure



Assessment philosophy

- Participation supports continuous learning.
- Midterms check individual foundations.
- The project tests integrated application and communication.

Midterm Coverage

Midterm 1: Foundations & Time Series

20%

Written examination, 75 minutes.

- Lecture 1: ML and Python
- Lecture 2: Data preparation
- Lecture 3: Financial time series

Midterm 2: Distributed Computing & Big Data ML

20%

Written examination, 75 minutes.

- Lecture 4: Big-data foundations
- Lectures 5–6: Spark and scaled ML
- Lecture 7: Advanced topics

Final Project: What Strong Work Looks Like

Technical quality

- Clear and valid dataset
- Reproducible data pipeline
- Appropriate scalable tools
- Defensible modeling and evaluation
- Organized source code

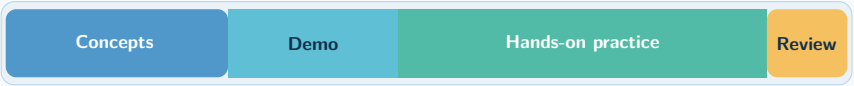
Financial and communication quality

- Relevant business question
- Correct financial interpretation
- Honest limitations and risks
- Actionable recommendation
- Clear presentation and Q&A

Project weight: 50%

Begin identifying a feasible topic and data source early; do not wait until the final stage.

A Typical 150-Minute Session



Illustrative format – timing changes with topic complexity

Theory-heavy sessions

More time is used to understand architectures, data movement, and modeling assumptions.

Application-heavy sessions

More time is reserved for coding, debugging, interpretation, and instructor feedback.

Technology Stack

Analysis: Python ▪ NumPy ▪ Pandas ▪ Scikit-learn

Efficient single-node: Vectorization ▪ Polars ▪ Lazy evaluation

Distributed computing: Apache Spark ▪ Spark SQL ▪ PySpark

Data and platforms: Parquet ▪ Cloud concepts ▪ Databricks

Scaled ML: Spark MLlib ▪ End-to-end analytical pipelines

How to Succeed in This Course

1. Review concepts before each practical session.
2. Run and modify every code example yourself.
3. Keep a record of errors and how you resolved them.
4. Interpret outputs in financial language.
5. Compare tools based on workload, not popularity.
6. Start the final project early and iterate.
7. Document assumptions, limitations, and risks.
8. Ask questions and contribute to class discussion.

Prepare
Before class

Practice
During class

Reflect
After class

Primary Course References: Books

[S1] Python for Finance

Hilpisch, Y. (2018). *Python for finance: Mastering data-driven finance* (2nd ed.). O'Reilly Media.

[S2] Advances in Financial Machine Learning

López de Prado, M. (2018). *Advances in financial machine learning*. John Wiley & Sons.

[S3] Data Preparation for Machine Learning

Brownlee, J. (2020). *Data preparation for machine learning: Data cleaning, feature selection, and data transforms in Python*. Machine Learning Mastery.

[Spark] Spark: The Definitive Guide

Chambers, B., & Zaharia, M. (2018). *Spark: The definitive guide—Big data processing made simple*. O'Reilly Media.

These books support the course's finance, machine-learning, data-preparation, and distributed-computing components.

Primary Course References: Online Documentation

[D1] Polars User Guide

Polars. (n.d.). *Polars user guide*.
https://pola-rs.github.io/polars-book/

[D2] Apache Spark Documentation

Apache Software Foundation. (n.d.). *Apache Spark official documentation*.
https://spark.apache.org/docs/latest/

[D3] Scikit-learn User Guide

Scikit-learn developers. (n.d.). *Scikit-learn user guide*.
https://scikit-learn.org/stable/user_guide.html

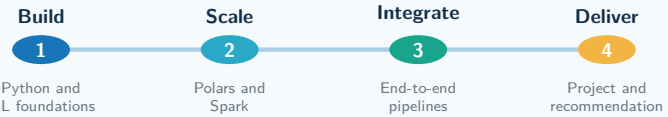
[D4] Databricks Documentation

Databricks. (n.d.). *Databricks documentation*.
https://docs.databricks.com/

Documentation habit

Confirm version-specific behavior in official documentation and test new APIs on small data before scaling up.

Your Course Roadmap



Remember

- Practice consistently
- Think financially
- Validate carefully
- Scale deliberately
- Communicate clearly

First action

Confirm that your Python environment and course notebooks run before the first technical lecture.

Welcome to the course!